Electrical power and energy

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Calculate electrical transfer rate and energy use.

SUCCESS CRITERIA - I can...

1. Use P=IV with watts, amps and volts.
2. Use E=Pt with seconds for joules.
3. Distinguish power from total energy.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

01 Warm-up: bring back an earlier idea

Two equal 10 Ω resistors are connected in parallel. Find equivalent resistance, then compare with a series connection of the same pair.

Marzano check - circle: ① ② ③

02 Worked example: explain the model

Reference: P = IV; E = PΔt; P = I^2R = V^2/R.
Constant power for E = PΔt; last equalities apply to an ohmic resistor.

Worked reasoning: A device at 6 V drawing 2 A receives power VI=12 W. Over 5 s it receives 60 J. Power is an energy-transfer rate.

Your turn: Draw an input-energy arrow and label voltage, current, power and elapsed time with distinct units.

Draw and label all energy transfers across the boundary.

Chosen system

Stored energy change: _____ J

Marzano check - circle: ① ② ③

03 Read and interpret the model

Two motors use the same energy but one takes half the time. Compare average power.

Draw and label all energy transfers across the boundary.

Chosen system

Stored energy change: _____ J

Marzano check - circle: ① ② ③

04 Apply the physics

A 12 V device draws a steady 0.75 A for 4 minutes. Calculate input power and total electrical energy in joules. Label the system model and show the time conversion.

Draw and label all energy transfers across the boundary.

Chosen system

Stored energy change: _____ J

Marzano check - circle: ① ② ③

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05 Explain your reasoning

A student says a 10 W motor always uses more energy than a 5 W motor. Supply a time comparison that disproves this.

Marzano check - circle: ① ② ③

06 Review: connect an earlier idea

A shallow-water model uses $v=\sqrt{(gh)}$. With g=10, h=40 m and travel distance 6 km, find speed and travel time.
State a model limitation.
Label a wave sketch to support the calculation or comparison.



Time or position: choose and label

Marzano check - circle: ① ② ③

07 Exit check

A 6 Ω resistor carries 2 A. Find its power using two equivalent equations.

Draw and label all energy transfers across the boundary.

Chosen system

Stored energy change: _____ J

Marzano check - circle: ① ② ③

08 STRETCH: test the model

A 12 V motor draws 1.5 A for 40 s and delivers 432 J of useful mechanical energy. Find electrical input energy, average useful power and efficiency.

Draw and label all energy transfers across the boundary.

Chosen system

Stored energy change: _____ J

Marzano check - circle: ① ② ③